SaberReal: Complete Research Reference for Vibra ATL 2026

No existing educational technology tool separates language proficiency from subject knowledge on assessments — making SaberReal a first-of-its-kind innovation addressing a problem that affects 5.3 million English Language Learners in U.S. public schools. Research consistently shows that standardized tests measure English fluency as much as content mastery, creating a phantom "achievement gap" that is largely a measurement artifact. For Georgia's 108,752 ELL students — tested exclusively in English with no native-language option (gbpi) — this means billions in misdirected remediation spending, tens of thousands of students misplaced in special education, and an entire generation of Hispanic students whose real knowledge remains invisible to the system that's supposed to serve them.

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1. The evidence is clear: tests measure English, not knowledge

The most important body of research for SaberReal comes from **Jamal Abedi**, the leading scholar on linguistic factors in assessment. His landmark 2001 study with Lord, published in *Applied Measurement in Education*, tested 1,174 eighth-graders (Taylor & Francis Online) across 11 Los Angeles-area schools (uga) and found that ELL students scored a mean of **11.56 out of 25 on math items vs. 15.14 for non-ELL students** (uga) (University of Georgia) — a gap driven substantially by the linguistic complexity of test items, not mathematical understanding. When test items were linguistically simplified (reducing passive voice, unfamiliar vocabulary, and complex sentence structures), scores improved by **3.7% for ELL students** compared to only 2.4% for non-ELLs. (uga) Critically, linguistic simplification helped ELLs **without inflating scores** for English-proficient students (ResearchGate) — proving the original wording was measuring language, not math. (Discovery Education)

Abedi's multi-site analysis for the U.S. Department of Education quantified this pattern across subjects. The ELL-to-non-ELL performance gap (measured as a "Disparity Index") was only **9.0% for math computation** — the lowest-language-demand content — but jumped to **26.4% for math problem solving** (ed) and **37% for 5th-grade science**, the highest-language-demand subjects. (ed) This gradient is the smoking gun: if ELL students truly lacked content knowledge, the gap would be consistent across formats. Instead, **the gap tracks almost perfectly with linguistic demand**, not content difficulty.

National Assessment data from NCES confirms the scale: **71% of 4th-grade English learners** scored "below basic" in reading vs. 30% of non-EL peers, and **43% of 4th-grade ELLs** scored below basic in math vs. 16% of non-EL peers. At 8th grade, **74% of ELLs** scored below basic in reading. (Vanderbilt Peabody) The foundational research by Carpenter et al. (1980) established that children perform **10% to 30% worse on arithmetic word problems than on comparable problems presented in numeric format** (uga) — a finding replicated consistently for four

decades. Robinson (2010) showed that kindergarten and first-grade Spanish-speaking ELs performed significantly better on mathematics assessments when tested in **Spanish instead of English**, with the effect strongest among low-SES students. PubMed Central

Perhaps most telling: Abedi found that **test reliability itself is compromised** for ELL students. Average test reliability (Cronbach's alpha) was **.603 for ELL students vs. .808 for non-ELL students** across ELA, science, and social science. Ed The tests are not just biased — they are literally less reliable instruments when used with ELL populations, producing noisier, less meaningful scores.

The performance gap also **widens with grade level**: the Disparity Index for reading was 23.4% in Grade 3, grew to 30.1% in Grade 4, and reached **41.1% by Grade 5**. ed As academic language becomes more complex, the measurement distortion worsens — meaning the students who most need accurate assessment are the ones whose scores are least trustworthy.

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2. Georgia tests 108,752 ELL students exclusively in English

Georgia is **one of only 19 states** that does not provide state standardized assessments in any language other than English — and only two other states with larger EL populations maintain this English-only policy. Georgia Budget and Policy I... gbpi The state's 108,752 ELL students (FY 2019), **78% of whom speak Spanish** at home, gbpi must take the Georgia Milestones Assessment entirely in English with limited accommodations. Colorín Colorado

The results are stark. On the 2023-24 Georgia Milestones 3rd-grade ELA assessment, **only 27.1% of Hispanic students** scored proficient or distinguished, Georgiareads compared to the state average of **38.4%** and **51.8% for White students**. English Learners specifically scored even lower: just **15.7% proficient**. Georgiareads georgiareads On the NAEP, only **3% of Georgia ELs scored proficient in 8th-grade reading** vs. 30% of non-ELs gbpi — a staggering **27-percentage-point gap**. Georgia Budget and Policy I... The hackathon team's reported 14-17 point gaps across subjects are consistent with state data and, if anything, conservative; the Migration Policy Institute found proficiency gaps between ELs and non-ELs of **19 to 39 percentage points** across Georgia Milestones EOC assessments.

Georgia's available ELL accommodations include extended time, small group testing, bilingual word-to-word glossaries (standard dictionaries only, not content-specific), and text-to-speech for math, science, and social studies. But these accommodations are marginal. Several Georgia districts provide more substantial language accommodations on classroom tests throughout the year, but **this same assistance is not allowed on Georgia Milestones** gbpi — meaning the state's high-stakes assessment systematically underestimates what ELL students actually know. Georgia Budget and Policy I...

Georgia's ELL population has grown **61% from FY 2011 to FY 2019** (Georgia Budget and Policy I…) — 3.5 times faster than the national average. (gbpi) Yet the state funds only about 1 dedicated ESOL instructor for every **39 ELL students**, (Georgia Budget and Policy I…) and only **1.2% of ESOL students** were enrolled in gifted programs vs. 11.6% statewide. Sixty-nine school districts with 10 or more ESOL students had **zero ELs in gifted programs**. (Georgia Budget and Policy I…) (gbpi) As the Georgia Budget and Policy Institute concluded, "testing students in a language other than their home language causes the dual harm of discouraging children who might otherwise be learning a great deal and withholding information from the state on the performance of these students." (Georgia Budget and Policy I…) (gbpi)

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3. Five million students, one systemic failure

The national scope of this problem is massive. As of fall 2021, **5.3 million students (10.6% of all K-12 enrollment)** were classified as English Learners in U.S. public schools, (Colorín Colorado) up from 4.6 million a decade earlier. (NCES) (NCES) **76.4% — approximately 4.0 million students — speak Spanish** as their home language. (NCES) (NCES) A critical and often-overlooked fact: **72% of ELL public school students ages 5-17 were born in the United States**. (Pew Research Center) These are overwhelmingly American children navigating an assessment system not designed for them.

The misidentification crisis represents perhaps the most damaging consequence. Research by Artiles, Rueda, Salazar, and Higareda (2005), published in *Exceptional Children*, found that ELLs in urban California districts were **27% more likely to be placed in special education in elementary grades and almost twice as likely in secondary grades**. (Waldenu) Students in English-only immersion programs were **almost three times as likely** to be placed in special education compared to ELs in bilingual programs by 12th grade. (ResearchGate) Sullivan's (2011) Arizona study documented a rising tide: the percentage of districts with ELL overrepresentation in special education grew from **32% to 42%** over eight years, (Waldenu) with ELLs most overrepresented in "specific learning disability" and "mild mental retardation" categories at risk ratios of **1.82 and 1.63** respectively. (ResearchGate)

In a particularly striking finding, Wilkinson, Ortiz, Robertson, and Kushner (2006) reviewed 21 ELLs identified as having learning disabilities and found that **10 — nearly half — were experiencing difficulties for reasons other than disability**, (PubMed Central) primarily language barriers. WIDA's analysis of federal data revealed that **45% of multilingual learners** dually identified under IDEA were classified as having a Specific Learning Disability, compared to about **35% of non-EL students** with disabilities (WIDA) — a 10-percentage-point disparity strongly suggesting language-based misidentification. The three root causes identified by Hamayan, Marler, Sanchez-Lopez, and Damico (2007) (Colorín Colorado) are problematic assessment practices conducted in English, the false assumption that disabilities can be reliably

identified across cultures, and a funding bias where schools receive additional per-pupil funding for special education placements. Colorín Colorado

The financial waste is substantial. The average cost per special education student is **$13,127/year** nationally (Bellwether, FY2020), Bellwether with California data showing costs of **$27,000/student for special ed vs. $10,000 for regular education** — nearly 3x the cost. ED100 Every ELL student incorrectly placed in special education due to language barriers being confused with learning disabilities costs an additional **$13,000-$17,000 per year** in misdirected spending.

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4. No existing tool does what SaberReal proposes

After comprehensive analysis of the EdTech landscape, the conclusion is definitive: **no existing commercial product provides an AI-powered diagnostic that simultaneously analyzes and separates whether a student's poor test performance stems from language proficiency barriers vs. actual subject knowledge gaps.** This is SaberReal's blue-ocean opportunity.

The closest existing tool is **Let's Go Learn's DORA Spanish EDELL diagnostic**, which allows educators to compare first-language Spanish reading skills to developing English reading skills. Let's Go Learn However, it is limited to reading comprehension only, is not AI-powered, and does not cover math, science, or social studies. Current best practice relies on teachers manually applying "differentiated scoring" rubrics — an entirely teacher-dependent process with no automation or scalability.

Here is how the major platforms fall short:

- **Khan Academy** offers Spanish-translated content but has no ELL-specific diagnostic features and no mechanism to separate language from knowledge. Its AI tutor Khanmigo operates in English only.
- **NWEA MAP Growth** is the most advanced bilingual adaptive testing platform, offering Spanish assessments in reading (K-8) and math (K-12) with cross-language score comparisons via IRT-based statistical linking. NWEA acknowledges that "with English language proficiency eliminated as a variable, Spanish-speaking students will have more opportunity to show what they know." NWEA However, it offers parallel assessments — not an automated diagnostic attributing performance differences to their root cause.
- **i-Ready** launched a Spanish reading diagnostic for K-6 in 2025-26 with authentically-written (not translated) Spanish items, Curriculum Associates but similarly does not cross-reference English proficiency with content knowledge.
- **WIDA ACCESS** measures English proficiency within content contexts ResearchGate but is a language proficiency test, not a content knowledge diagnostic.

- **Ellevation Education** serves 1,200+ districts and 2 million students as an EL program management platform but does not administer assessments or diagnose language vs. content gaps.
- **Imagine Learning** offers first-language support in 15 languages and reports that ELL users outperform non-participating peers on WIDA ACCESS by **38%**, but focuses on building skills rather than diagnostic separation.

In academic research, **PyrEval** — an NLP system documented in a 2026 paper in the *British Journal of Educational Technology* — is the closest proof of concept. PyrEval automates content assessment of scientific explanations **without linguistic discrimination**, evaluating what students know about science rather than how they express it in English. It specifically did not penalize essays with non-normative linguistic features. Wiley Online Library This demonstrates that AI **can** separate content knowledge from language quality, but PyrEval remains an academic tool, not a commercial product.

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5. A $53-420 million market with strong technical foundations

The business case for SaberReal rests on three pillars: the scale of wasted spending, the size of the addressable market, and the funding streams available to schools.

The wasted spending argument is the most emotionally and financially compelling pitch element. Title III — the only federal funding stream specifically for ELL programs Ed — provides approximately **$890 million/year**, Edweek amounting to roughly **$168 per ELL student** — widely considered insufficient. Title I provides **$18.3 billion annually** to support low-income schools, New America U.S. Department of Educati... with ELL accountability now embedded under ESSA. Under ESSA, schools where ELL subgroups underperform risk identification for Comprehensive Support and Improvement (CSI) or Targeted Support and Improvement (TSI), triggering mandatory improvement plans and resource allocation reviews. Congress.gov Migration Policy Institute Meanwhile, McKinsey's research estimates that the racial achievement gap costs the U.S. economy **$426-705 billion per year** in lost GDP Cuny — what they called "the economic equivalent of a permanent, deep recession." Children's Defense Fund

The market opportunity is well-defined. The U.S. K-12 testing and assessment market is valued at **$10.95-16.2 billion** (2025) and projected to grow to **$15.7-32 billion by 2030-2033**, with adaptive assessment as the fastest-growing segment at **9.1% CAGR**. Assessment platform pricing benchmarks at **$10-15 per student per year** (NWEA MAP Growth lists at $14.50/student). amazonaws SaberReal's Total Addressable Market ranges from **$53-80 million** at assessment-only pricing (5.3M ELLs × $10-15) to **$210-420 million** if targeting all Hispanic/Latino students (~14M) at diagnostic-plus-remediation pricing ($15-30/student). A realistic Year 1-2 target of 1-3% SAM penetration yields **$420K-2.1 million** in revenue.

Schools have multiple funding sources to purchase SaberReal: Title III funds ($890M/year specifically for supplemental ELL services), Education Law Center Edweek Title I funds ($18.3B/year for academic improvement), IDEA funds (savings from reducing special ed misplacements at $13,000-17,000/student/year), state ELL supplemental funding, and district general assessment budgets.

The technical feasibility is strong. Modern LLMs have robust multilingual capabilities in English and Spanish. The PyrEval research proves NLP can assess content without linguistic discrimination. Wiley Online Library Auto Error Analyzer (Cambridge, 2025) demonstrates that Llama 3.3 can detect L2 language errors with **.94 correlation to human raters**. Cambridge Core QueryQuilt (arXiv, 2024) shows LLMs can identify knowledge gaps from student interactions with **100% accuracy on simulated students and 95% completeness on real data**. IRT-based Computer Adaptive Testing is mature technology already deployed by MAP Growth NWEA and i-Ready for cross-language score linking. A viable SaberReal architecture would combine dual-language adaptive assessment (IRT/CAT), NLP error classification distinguishing language-based from knowledge-based errors, and LLM-powered analysis of free responses in both languages.

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6. Decades of legal precedent demand this solution

The legal framework supporting SaberReal's mission is extensive and deeply established. **Lau v. Nichols** (1974) saw the Supreme Court unanimously rule that identical education does not constitute equal education — school districts receiving federal funds must take affirmative steps to overcome language barriers. Wikipedia +4 **Castañeda v. Pickard** (1981) established the three-prong test still used today to evaluate ELL programs: Springer the program must be based on sound educational theory, adequately implemented, and produce results. PubMed Central University of Nebraska-Lin... **Plyler v. Doe** (1982) established that states cannot deny education to undocumented children. The 74 Million Wikipedia The **Equal Educational Opportunities Act of 1974** explicitly prohibits denying equal educational opportunity through failure to overcome language barriers. Study.com +2

Under **ESSA** (2015), states must include all ELs in annual standardized testing Center for Parent Informati... and must "**make every effort**" to develop assessments in languages present to a significant extent in their student populations New America — a provision Georgia has not fulfilled despite having over 108,000 ELLs. ESSA also requires ELs as a mandatory accountability subgroup, Migration Policy Institute meaning schools are financially and reputationally penalized when ELL students underperform Congress.gov — but the system provides no mechanism to determine whether that underperformance reflects actual knowledge gaps or measurement artifacts from linguistic bias.

The **Individuals with Disabilities Education Act (IDEA)** requires that evaluations be conducted in a child's native language, using varied assessment tools, free from racial or cultural discrimination. VIVA's Pressbooks Colorín Colorado The 2015 OCR/DOJ Dear Colleague Letter explicitly states that schools must evaluate ELs suspected of having a disability "in an appropriate language based on the student's needs" to avoid misidentification. Yet systematic violations persist: IDEA 2004 specifically acknowledged the disproportionate number of culturally and linguistically diverse students in special education. Leadersproject ResearchGate

The current policy environment creates urgency. The Trump administration proposed **eliminating the entire $890 million Title III program** Education Week in its FY2026 budget, Education Law Center temporarily impounded these funds, rescinded Obama-era ELL guidance, and designated English as the official language via executive order. Education Law Center While the underlying statutory obligations remain in effect, reduced federal oversight increases the need for tools like SaberReal that can help schools meet their legal obligations efficiently with fewer resources.

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7. What reclassified students reveal about hidden potential

The most powerful proof point for SaberReal's thesis comes from reclassification data. When ELL students are reclassified as English-proficient, their academic performance tells a dramatic story. The Public Policy Institute of California's longitudinal analysis found that reclassified former EL (RFEP) students "not only outperform EL students, but also often do as well as — or better than — native English speakers" on standardized tests. PPIC In 2011, **94% of former ELs** passed California's High School Exit Exam ELA portion vs. **87% of English-only students** and only **44% of current ELs**. PPIC A WestEd analysis found that students scoring a 4 on the ELPAC "outperformed peers who only spoke English at home with similar levels of parent education and family income." EdSource

This paradox — where the same students perform at or above grade level once their English improves — is exactly what SaberReal is designed to reveal *before* years of misplacement, stigma, and wasted resources. The NCTM's official position captures it succinctly: "ELLs commonly know more math than they can demonstrate on traditional assessment instruments." Discovery Education

Research by Hakuta, Butler, and Witt (2000) established that it takes **5 to 7+ years** for most ELLs to gain sufficient mastery of academic English to match English-speaking peers on assessments ed — meaning years of standardized test data may systematically misrepresent these students' actual capabilities. During those years, students face compounding consequences: placement in remedial rather than grade-level courses, exclusion from gifted programs (only 1.2% of Georgia ESOL students vs. 11.6% statewide), gbpi potential misidentification for special education, Continental Press and erosion of academic confidence. Georgia Budget and Policy I...

Texas's experience with bilingual programs offers a systemic counterpoint. Texas enrolls approximately 40% of its ELs in bilingual education — twice California's rate (The 74 Million) — and offers the STAAR test in Spanish for grades 3-5 in reading, math, writing, and science. The reading/language arts assessments are independently developed in Spanish, not simply translated. (Texas Assessment) Texas's bilingual ELL students outperform California's EL counterparts on multiple metrics, (The 74 Million) providing evidence that native-language assessment and instruction produce measurably better outcomes.

Conclusion: SaberReal fills a gap that decades of research have documented

The research paints an unambiguous picture. The "achievement gap" for Hispanic ELL students is substantially a **measurement gap** — standardized tests conflate English proficiency with content knowledge, (Semantic Scholar) and the distortion worsens as academic language complexity increases. Nearly half of ELL special education referrals may be misidentifications rooted in language barriers, (PubMed Central) (Continental Press) costing $13,000-17,000 per misplaced student annually. Georgia's English-only assessment policy compounds the problem for its rapidly growing ELL population. (gbpi) Yet despite 50 years of legal precedent demanding equitable assessment, 40 years of research documenting linguistic bias, and a $10+ billion testing market — **no commercial tool exists that provides the core diagnostic SaberReal proposes**: an AI-powered separation of language barriers from knowledge gaps.

The novel insight SaberReal brings is not just bilingual testing (MAP Growth and i-Ready already offer that) but **automated causal attribution** — using NLP and adaptive assessment to tell teachers, parents, and administrators *why* a student scored poorly. The technology is proven in adjacent research contexts. The market is large and growing. The legal mandate is clear. The funding streams exist. And **4 million Spanish-speaking students are waiting** (NCES) for an assessment system that finally sees what they know, not just how they say it.

Appendix: Key statistics for the pitch deck

Statistic	Value	Source
ELL students in U.S. public schools	**5.3 million** (10.6% of K-12)	NCES, Fall 2021
Spanish-speaking ELLs	**4.0 million** (76.4%)	NCES, Fall 2021
ELLs born in the United States	**72%**	Census/Pew Research
Georgia ELL students	**108,752**	GOSA, FY2019

Statistic	Value	Source
Georgia ELL population growth (2011-2019)	**+61%** (3.5x national avg)	GBPI
GA 3rd-grade ELA: Hispanic proficiency	**27.1%** vs. 38.4% state avg	GaDOE, 2023-24
GA 3rd-grade ELA: English Learner proficiency	**15.7%**	GaDOE, 2023-24
GA 8th-grade reading: EL proficiency (NAEP)	**3%** vs. 30% non-EL	NAEP, 2019
4th-grade ELLs below basic in reading nationally	**71%**	NCES
Math performance gap: computation vs. problem solving	**9% vs. 26.4%** (DI)	Abedi & Sato, U.S. DOE
ELL score improvement with linguistic simplification	**+3.7%** (vs. +2.4% non-ELL)	Abedi & Lord, 2001
ELL test reliability vs. non-ELL	**.603 vs. .808** (alpha)	Abedi & Sato
ELLs misidentified as learning disabled	**~50% of referrals**	Wilkinson et al., 2006
Special ed cost per student (national avg)	**$13,127/year**	Bellwether, FY2020
Special ed cost premium (California)	**$27K vs. $10K** regular ed	California LAO, 2019
GA ESOL students in gifted programs	**1.2%** vs. 11.6% statewide	GaDOE, FY2019
States without non-English assessments	**19** (including Georgia)	GBPI/MPI
Title III federal ELL funding	**$890 million/year**	U.S. DOE
Title I funding	**$18.3 billion/year**	U.S. DOE
K-12 assessment market	**$10.95-16.2 billion**	Mordor/MFRF, 2025
Achievement gap GDP cost	**$426-705 billion/year**	McKinsey, 2020
Existing tools separating language from knowledge	**Zero**	Competitive analysis
Years for ELLs to gain academic English proficiency	**5-7+ years**	Hakuta, Butler & Witt, 2000
Reclassified ELs passing CA exit exam (ELA)	**94%** vs. 87% English-only	PPIC